StartSmart – A Startup Feasibility & Resource Planning Platform

Software Requirements Specification

Prepared by

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the requirements for developing **StartSmart**, a web-based startup feasibility and resource planning platform.

The main objective of StartSmart is to help aspiring entrepreneurs evaluate the feasibility of their startup ideas by analyzing budget, industry type, estimated expenses, risks, and expected profit. The system provides detailed cost breakdowns, marketing insights, influencer contacts, raw material information, and business roadmap guidance to enable informed decision-making.

1.2 Scope

StartSmart will:

- Allow users to input their startup budget.
- Allow users to select a business industry (e.g., clothing, food, cosmetics, YouTube).
- Calculate startup feasibility based on estimated costs.
- Provide profit estimation and break-even analysis.
- Generate SWOT and risk analysis reports.

- Recommend funding sources and alternative business ideas.
- Export a downloadable business plan report (PDF).

The platform will be accessible through web browsers and built using modern web technologies.

1.3 Definitions, Acronyms, and Abbreviations

- MERN: MongoDB, Express.js, React.js, Node.js
- SWOT: Strengths, Weaknesses, Opportunities, Threats
- ROI: Return on Investment
- PDF: Portable Document Format
- RBAC: Role-Based Access Control

1.4 References

Technology Documentation:

- MongoDB Official Documentation
- Express.js Documentation
- React.js Documentation
- Node.js Documentation

Startup Analysis Concepts:

- Break-Even Analysis

- SWOT Analysis Framework
- Risk Assessment Models

1.5 Overview

Section 2 describes the overall system and its users.
Section 3 specifies functional and non-functional requirements.
Section 4 explains the technical architecture.
Section 5 presents the Agile development plan.
Section 6 defines acceptance criteria.

2. Overall Description

2.1 Product Perspective

- StartSmart is a standalone web-based startup feasibility and planning platform developed using the MERN stack (MongoDB, Express.js, React.js, Node.js). The system operates as a client-server application where users interact with a responsive front-end interface, and all calculations and business logic are handled by the backend server.
- The platform integrates internal cost models, startup industry data, and analytics tools to provide intelligent business feasibility insights. Future versions may integrate third-party APIs such as funding databases, market analytics tools, or financial APIs for more accurate real-time estimations.

2. Overall Description

2.2 Product Features

1. **Budget Input Module**: Users enter their available startup investment amount. This budget acts as the primary financial input for feasibility calculation and business planning.
2. **Industry Selection Module:** Users select business categories such as clothing, food, YouTube, or cosmetics. The system loads predefined cost structures and market assumptions based on the selected industry.
3. **Startup Feasibility Calculator:** Calculates the total estimated startup cost and determines whether the selected business idea is financially feasible within the user's budget.
4. **Expense Breakdown Engine:** Displays a detailed distribution of costs including equipment, raw materials, marketing, operations, licensing, and miscellaneous expenses.
5. **Influencer Recommendation System:** Shows relevant influencer categories, estimated promotional charges, and potential marketing reach to support business promotion planning.
6. **Raw Material Information Module:** Displays required raw materials, supplier categories, and essential operational inputs for the selected business type.
7. **Profit Estimation System:** Estimates expected revenue and projected profit based on industry benchmarks and financial assumptions.
8. **Startup Recommendation Engine:** Suggests alternative startup ideas if the selected business is not feasible within the provided budget.
9. **Feasibility Result Dashboard:** Shows overall feasibility status, total estimated cost, expected profit, and any budget gap.
10. **Startup Roadmap Generator:** Provides a step-by-step startup guide including planning, procurement, marketing launch, and operational setup.
11. **Admin Dashboard:** Allows administrators to manage industry data, cost models, influencer records, and system parameters.
12. **Analytics Module:** Shows visual analysis of costs, revenue projections, and profit trends using charts and graphs.
13. **Resource Requirement Module:** Displays required equipment, tools, services, and other operational resources needed for business setup.

14. **Database Management System:** Stores user data, industry information, financial models, and analysis results securely.
15. **Risk Assessment Module:** Evaluates potential business risks such as market competition, demand uncertainty, and capital limitations.
16. **Break-Even Analysis Module:** Calculates the break-even point based on cost and projected revenue estimation.
17. **Funding Recommendation Module:** Suggests suitable funding sources such as bank loans, investors, crowdfunding, or partnerships.
18. **SWOT Analysis Tool:** Generates strengths, weaknesses, opportunities, and threats analysis for the selected business type.
19. **Marketing Budget Planner:** Estimates required marketing expenses based on industry type and promotional strategy.
20. **Business Plan Export Module:** Generates a downloadable business plan report in PDF format containing feasibility results and financial analysis.

2.3 User Classes and Characteristics

Entrepreneurs: Individuals planning to start a business. They may not have advanced financial knowledge and require a simple, user-friendly interface with clear analysis results.

Administrators: System managers responsible for maintaining industry cost data, managing system parameters, and monitoring analytics. They require full access to backend management tools.

2.4 Operating Environment

Web Application: Accessible on modern web browsers (Chrome, Firefox, Edge, Safari).

Mobile Compatibility: Responsive design compatible with smartphones and tablets.

Server-Side: Node.js and Express.js running in a cloud-based or containerized environment.

Database: MongoDB or MySQL database for secure data storage.

Hosting: Deployment on cloud platforms such as AWS, Azure, Google Cloud, or equivalent hosting services.

2.5 Constraints

Data Accuracy: Feasibility results depend on predefined industry cost models.

Security: The system must implement HTTPS and secure database storage.

Performance: The system should provide fast calculation and dashboard response time.

Scalability: The architecture should support increasing users without major redesign.

Time & Resource: The project will be developed within 8–9 weeks by a small development team.

2.6 Assumptions and Dependencies

Users provide accurate budget information.

Industry cost data is properly configured by administrators.

Users have a stable internet connection.

PDF generation libraries and database services function reliably.